

Kindle Agent - System Architecture Document

1. High-Level Architecture

User ↓ Frontend ↓ Backend API ↓ Agent Layer ↓ MCP Server ↓ Document Tools ↓ External Utilities

2. Component Breakdown

Frontend

Responsibilities:

- File Upload
- Progress Display
- Metadata Editing
- File Download

Technology:

- React
 - TailwindCSS
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Backend API

Responsibilities:

- Receive uploads
- Store files
- Trigger agent workflow
- Return results

Technology:

- FastAPI
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Agent Layer

Responsibilities:

- Decide workflow
- Select tools

- Manage execution sequence

Example:

PDF Upload ↓ Analyze PDF ↓ Need OCR? ↓ Run OCR ↓ Convert Format ↓ Return Output

MCP Server

Responsibilities:

Expose tool interfaces.

Tools:

- analyze_pdf
 - run_ocr
 - extract_metadata
 - convert_document
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Tool Layer

Implements actual business logic.

Examples:

analyze_pdf.py run_ocr.py metadata.py convert.py

External Utilities

OCR:

- Tesseract OCR

Conversion:

- Calibre
- ebook-convert CLI

PDF Analysis:

- PyMuPDF
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3. Data Flow

Upload PDF ↓ Store File ↓ Agent Receives Task ↓ Agent Calls MCP Tool ↓ Tool Processes File ↓ Result Returned ↓ Agent Continues Workflow ↓ Output Generated ↓ Download Link Returned

4. Scalability Considerations

Future:

Frontend ↓ FastAPI ↓ Message Queue ↓ Agent Workers ↓ MCP Servers ↓ Storage

This allows processing many PDFs simultaneously.